**PPA 4**

**Title**

# Physical Substrates for Transformation-First Computation

## Field of the Invention

The invention relates to computational hardware, silicon architectures, physical computing systems, and unconventional computing substrates.

## Background

Existing hardware architectures implement computation by executing instructions, evaluating logic gates, and stepping through time-based cycles.

These architectures inherently enforce linear evaluation and symbolic abstraction.

## Summary of the Invention

The invention describes physical substrates—implemented in silicon or other materials—in which:

- Spatial configuration performs computation
- Transformation precedes representation
- Logical behavior arises from physical constraint

No clocked instruction execution is required.

## Detailed Description

The substrate may consist of spatially arranged elements whose interactions enforce symmetry, adjacency, and transformation constraints.

Computation is the physical evolution of the system itself.

## Key Novel Claims (Provisional)

- Hardware that computes through physical transformation
- Elimination of instruction execution
- Spatial logic embedded in material structure
- Chips implementing logic via constraint rather than gates